



BITS Pilani

Pilani Campus

Mathematical Foundations

MFDS & MFML Team



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Mathematical Foundations

Webinar#1

Agenda



Problems on

1. **Solution to the system of Linear Equations**
2. **Vector space**
3. **Subspace**
4. **Linearly Independent and Dependent Vectors**
5. **Basis and Dimension**
6. **Row Space and Null Space**
7. **Inner Product**
8. **Symmetric Matrices and Positive-Definite Matrices**

Some Important Definitions

Elementary row transformation

Example: $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

Sl. No.	Elementary row transformation	Notation	Resultant of a matrix A
1	Interchange any two rows Here first and 2 nd rows are interchanged	$R_1 \leftrightarrow R_2$	$\begin{bmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$
2	Multiplication of a row by a non zero constant Here 3 rd row is multiplied by k	kR_3	$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ kc_1 & kc_2 & kc_3 \end{bmatrix}$
3	Add to a row k times another row Here k times first row is added to 2 nd row	$R_2 \rightarrow kR_1 + R_2$	$\begin{bmatrix} a_1 & a_2 & a_3 \\ ka_1 + b_1 & ka_2 + b_2 & ka_3 + b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

Echelon Form and Reduced Row Echelon Form

A rectangular matrix is in echelon form (or row echelon form) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

If a matrix in echelon form satisfies the following additional conditions, then it is in reduced echelon form (or reduced row echelon form):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

Rank of a matrix:

The rank of a matrix A in its echelon form is equal to the number of non zero rows. It is denoted by $\rho(A)$.

Find the rank of the following Matrices:

$$\begin{bmatrix} 1 & 0 & 5 & 0 & 0 & 8 \\ 0 & 1 & 7 & 0 & 0 & 9 \\ 0 & 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 4 \\ 0 & 0 & 1 & 2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. System of linear equations and Consistency

Consider a system of 'm' linear equations in 'n' unknowns as follows

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\dots \dots \dots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Where a_{ij} 's and b_i 's are constants.

- If $b_1, b_2, \dots, b_m$ are all zero, the system is said to be **homogeneous**. Otherwise, it is said to be **Non-homogeneous**.
- The set of values $x_1, x_2, \dots, x_n$ which satisfy all the equations simultaneously is called a solution of the system of equations.
- A system of linear equations is said to be **consistent** if it possess a solution. Otherwise, it is said to be **inconsistent**.

2. Vector Space

Let R denote the set of all real numbers (Scalars), V be a non-empty set for every ordered pair on which two operations (vector addition and scalar multiplication) are defined. If the listed axioms are satisfied for every u, v, w in V and scalars c and d in F , then V is called a vector space over the field.

- If for all u, v in V then $u + v$ is a vector in V (Closure under addition).
- If for all u, v in V then $u + v = v + u$ (Commutative property of addition).
- If for all u, v, w in V then $(u + v) + w = u + (v + w)$ (Associative property of addition).
- If for all u in V , there is a zero vector 0 in V such that, we have $(u + 0) = u = (0 + u)$ (Additive identity).
- If for all u in V , there is a vector in V denoted by $-u$ such that $u + (-u) = 0 = (-u) + u$ (Additive inverse).
- If for all c in F and u in V then cu is in V (Closure under scalar multiplication).
- If for all c in F and u in V then $c(u + v) = cu + cv$ (Distributive property of scalar multiplication).
- If for all c, d in F and u in V then $(c + d)u = cu + du$ (Distributive property of scalar multiplication).
- If for all c, d in F and u in V then $c(du) = (cd)u$ (Associate property of scalar multiplication).
- If for all u in V and 1 in F then $1(u) = u$ (Scalar identity property).

3.Subspace



A non-empty subset W of a vector space V is a subspace of V if and only if the following conditions are satisfied:

W is non empty.

Closure under Addition: For all $\alpha, \beta \in W$, $\alpha + \beta \in W$.

Closure under Scalar Multiplication: For all $c \in F$ and $\alpha \in W$, $c \cdot \alpha \in W$.

Example: Let V be the set of all ordered pairs (x, y) where x and y are real numbers. Let $u = (x_1, y_1)$ and $v = (x_2, y_2)$ be two elements in V . Define the addition $u + v = (x_1, y_1) + (x_2, y_2) = (2x_1 - 3x_2, y_1 - y_2)$ and the scalar multiplication as $k(x_1, y_1) = ((kx_1)/3, (ky_1)/3)$. Show that $+$ in V does not satisfy commutative and Associative laws. Also the scalar multiplication does not satisfy the axiom 1. $\mathbf{v} = \mathbf{v}$.

Solution:

- For u, v in R^2 ,

$$\begin{aligned} u + v &= (x_1, y_1) + (x_2, y_2) = (2x_1 - 3x_2, y_1 - y_2) \neq (2x_2 - 3x_1, y_2 - y_1) \\ &\neq (x_2, y_2) + (x_1, y_1) \neq v + u. \end{aligned}$$

which does not satisfy the commutative law.

- For u, v, w in R^2 ,

$$\begin{aligned}
 \text{Consider } (u+v)+w &= \{(x_1, y_1) + (x_2, y_2)\} + (x_3, y_3) \\
 &= (2x_1 - 3x_2, y_1 - y_2) + (x_3, y_3) \\
 &= (2(2x_1 - 3x_2) - 3x_3, y_1 - y_2 - y_3) \\
 &= (4x_1 - 6x_2 - 3x_3, y_1 - y_2 - y_3)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now consider } u + (v+w) &= (x_1, y_1) + \{(x_2, y_2) + (x_3, y_3)\} \\
 &= (x_1, y_1) + (2x_2 - 3x_3, y_2 - y_3) \\
 &= (2x_1 - 3(2x_2 - 3x_3), y_1 - (y_2 - y_3)) \\
 &= (2x_1 - 6x_2 + 9x_3, y_1 - y_2 + y_3)
 \end{aligned}$$

Thus, $(4x_1 - 6x_2 - 3x_3, y_1 - y_2 - y_3) \neq (2x_1 - 6x_2 + 9x_3, y_1 - y_2 + y_3)$.

which does not satisfy the Associative law

- For u in R^2 1 in F ,

$$\text{we have } 1 \cdot u = 1 \cdot (x_1, y_1) = ((x_1/3), (y_1/3)) \neq (x_1, y_1) \neq u$$



4. Linearly Independent and Dependent Vectors

Linear Combination :

Let V be a vector space and $u_1, u_2, \dots, u_n$ be any n vectors of V . the vector of the form $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = \alpha$ where $\alpha_1, \alpha_2, \dots, \alpha_n$ in F is called a linear combination of the vectors $u_1, u_2, \dots, u_n$.

Linear Span :

Let V be a vector space and let $S = \{u_1, u_2, \dots, u_n\}$ be a non-empty subset of V . The linear span of S is the set defined by $L(S) = \{\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n : \alpha_i \in F, 1 \leq i \leq n\}$.

Linearly Dependent Vectors

A set $\{v_1, v_2, \dots, v_n\}$ of vectors of a vector space V is said to be linearly dependent if there exist scalars $c_1, c_2, \dots, c_n$ in R , not all zeros, such that $c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$.

Linearly Independent Vectors:

A set $\{v_1, v_2, \dots, v_n\}$ of vectors of a vector space $V(F)$ is said to be linearly independent if and only if $c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$ implies $c_1 = 0, c_2 = 0, \dots, c_n = 0$.



5. Basis and Dimension

Basis

A non-empty subset B of a vector space V is called a basis of V if

- B is a linearly independent set, and
- $L(B) = V$, i.e., every vector in V can be expressed as a linear combination of the elements of B .

Dimension

The dimension of a finite dimensional vector space V is the number of elements in any basis of V and is denoted by $\dim(V)$.



6. Row Space and Null Space

Row Space

- The row space of an m by n matrix A is the subspace of $\mathbb{R}^n$ spanned by its row vectors.

Null Space

The null space (or kernel) is the set of all vectors x in $\mathbb{R}^n$ that satisfy the equation $Ax = 0$.

Problem 1



$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

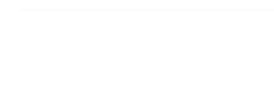
(A) Using elementary row operations, write the matrix in its row echelon form



(B) Let $\mathbf{V}$ be a vector subspace spanned by the columns of matrix $\mathbf{A}$. Find the basis and dimension of $\mathbf{V}$.



(C) Let $\mathbf{V}$ be a vector subspace spanned by vectors $\mathbf{x}$, such that $\mathbf{Ax} = \mathbf{0}$. Find the basis and dimension of $\mathbf{V}$.



Problem 1 –Solution



(A)

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3 * R_1, R_3 \leftarrow R_3 - 2 * R_1, R_2 \leftarrow R_2 / 5, R_3 \leftarrow R_3 - R_2, R_1 \leftarrow R_1 - 2 * R_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

NOTE: REF is not unique.

(B) column number 1 and column number 3 have pivot.

Basis of column space is:

$$\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

Rank of col space = dimension of $\mathbf{V} = 2$

As we have seen, the echelon form is $\begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Converting back to homogeneous equations:

$$x_1 - 2x_2 + 2x_3 + 3x_4 - x_5 = 0 \quad (\text{same equation})$$

$$x_3 + 2x_4 - 2x_5 = 0$$

(2nd row gives the same equation),

Since x_2, x_4, x_5 are free variables,

let us assign: $x_2 = r, \quad x_4 = s, \quad x_5 = t$

Since $x_3 + 2x_4 - 2x_5 = 0 \Rightarrow x_3 = -2x_4 + 2x_5 = -2s + 2t$

Now $x_1 - 2x_2 + 2x_3 + 3x_4 - x_5 = 0$

$$\Rightarrow x_1 = 2x_2 - 2x_3 - 3x_4 + x_5$$

$$= 2r - 2(-2s + 2t) - 3s + t$$

$$= 2r + s - 3t$$

The solution of $Ax = 0$ is

$$\begin{bmatrix} 2r + s - 3t \\ r \\ -2s + 2t \\ s \\ t \end{bmatrix} = r \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

(C) From REF, the null space is:

$$\begin{bmatrix} x \\ r \\ y \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} t$$

basis =

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

Rank of null space = dimension of $\mathbf{V} = 3$

Problem 2

innovate

achieve

lead

- (A) Suppose $A = [C_1, C_2, C_3, C_4]$ where $C_m \in \mathbb{R}^4, m = 1, 2, 3, 4$ are columns of A . It is known that $\text{rank}(A) = 2$ and $C_2 = 3C_1$ and $C_4 = 2C_1 + 3C_3$. If a particular solution of $Ax = b$ is $[1, 0, 1, 0]^T$, then
- (i) Find the general solution of $Ax = b$.
 - (ii) Find b if $\text{RREF}(A) = A$.
- (B) Consider $M = \{A \in \mathbb{R}^{2 \times 2} \mid A = -A^T\}$.
- (i) Prove that M is subspace of vector space V , where V is set of all 2×2 real matrices.
 - (ii) Prove or disprove $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for M defined in (i).
 - (iii) Prove or disprove that $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} \right\}$ is a linearly independent set.

Problem 2 – Solution

(A) Now $A = [C_1, C_2, C_3, C_4]$ where $C_m \in \mathbb{R}^4, m = 1, 2, 3, 4$ are columns of A . It is known that $\text{rank}(A) = 2$

$$(i) C_2 = 3C_1 \Rightarrow 3C_1 - C_2 + 0C_3 + 0C_4 = 0$$

$$\Rightarrow [C_1, C_2, C_3, C_4] \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$C_4 = 2C_1 + 3C_3 \Rightarrow 2C_1 + 0C_2 + 3C_3 - 1C_4 = 0$$

$$\Rightarrow [C_1, C_2, C_3, C_4] \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus, the general solution of $Ax = b$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} \mid \forall \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

(ii) Now $A = [C_1, C_2, C_3, C_4] = [C_1, 3C_1, C_3, 2C_1 + 3C_3]$
and $\text{rank}(A) = 2$.

Therefore C_1, C_3 are pivot columns and $\text{RREF}(A) = A$

$$\Rightarrow C_1 = [1, 0, 0, 0]^T \text{ and } C_3 = [0, 1, 0, 0]^T$$

$$\Rightarrow C_2 = [3, 0, 0, 0]^T \text{ and } C_4 = [2, 3, 0, 0]^T$$

$$\Rightarrow A = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Therefore } b = A[1, 0, 1, 0]^T = [1, 1, 0, 0]^T$$

(B) (i) Clearly the zero matrix of order 2, $0 \in M \Rightarrow M \neq \phi$.

Let $A, B \in M, k \in \mathbb{R}$

$$\Rightarrow A = -A^T, B = -B^T$$

$$\Rightarrow A + B = -A^T - B^T = -(A + B)^T$$

$$\Rightarrow A + B \in M.$$

$$kA = k(-A^T) = (-k)A^T = -(kA)^T$$

So, M is a subspace.

(ii) Clearly $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ does not belong to M and hence cannot be a basis of M .

(Kindly award full marks for any valid reason for disproving the claim.)

(iii) Consider $a \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + b \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix} + c \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$\Rightarrow a + 5b = 0$$

$$-a - 5b - 3c = 0$$

$$a + 5b + 3c = 0$$

$$5b + c = 0$$

$\Rightarrow a = b = c = 0$. Hence it is a linearly independent set.

Problem 3



Consider the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 7 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

- (A) Using elementary row operations, write the matrix in its row echelon form
- (B) Let $\mathbf{V}$ be a vector subspace spanned by the columns of matrix $\mathbf{A}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis set?
- (C) Let $\mathbf{V}$ be a vector subspace spanned by vectors $\mathbf{x}$, such that $\mathbf{Ax} = \mathbf{0}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis set?
- (D) Give the set of linearly independent rows of $\mathbf{A}$. What is the number of vectors in this set?

Problem 3 Solution



(A)

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 7 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{6}{11} \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \leftarrow R_1/2, R_2 \leftarrow R_2 - 3 * R_1, R_3 \leftarrow R_3 - 3 * R_1,$$

$$R_2 \leftarrow R_2/(11/2), R_3 \leftarrow R_3/(-1/2), R_3 \leftarrow R_3 - R_2 \rightarrow$$

NOTE: REF is not unique.

(B) All columns have pivot.

All columns form the basis of column space

$$\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

Rank of col space = 3

(C) From REF, the null space is: $\{0\}$
Rank of null space = 0

(D) Since each column has a pivot,
A has 3 linearly independent rows.

The set of Linearly Independent
rows is given below.

There are three elements

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

7. Norms and Inner Product



Norms

- ▶ A norm on a vector space is a function $\|\cdot\| : V \rightarrow \mathbb{R}$, $\mathbf{x} \rightarrow \|\mathbf{x}\|$ which assigns to each vector $\mathbf{x}$ a length $\|\mathbf{x}\|$ such that for all $\lambda \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in V$ the following properties hold:

- ▶ Absolutely homogeneous: $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$
- ▶ Triangle inequality: $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
- ▶ Positive definite: $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0 \implies \mathbf{x} = 0$

- ▶ Manhattan norm : $\|\mathbf{x}\| = \sum_{i=1}^{i=n} |x_i|$

- ▶ Euclidean norm : $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{i=n} x_i^2}$.

Inner Product

- ▶ Dot product in $\mathbb{R}^n$ is given by $\mathbf{x}^T \mathbf{y} = \sum_{i=1}^{i=n} x_i y_i$
- ▶ A bilinear mapping Ω is a mapping with two arguments and is linear in both arguments: Let V be a vector space such that $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, and let $\lambda, \psi \in \mathbb{R}$. Then we have $\Omega(\lambda \mathbf{x} + \psi \mathbf{y}, \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{z}) + \psi \Omega(\mathbf{y}, \mathbf{z})$, and $\Omega(\mathbf{x}, \lambda \mathbf{y} + \psi \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{y}) + \psi \Omega(\mathbf{x}, \mathbf{z})$.
- ▶ Let V be a vector space and $\Omega : V \times V \rightarrow \mathbb{R}$ be a bilinear mapping that takes two vectors as arguments and returns a real number. Then Ω is called symmetric if $\Omega(\mathbf{x}, \mathbf{y}) = \Omega(\mathbf{y}, \mathbf{x})$. Also Ω is called positive-definite if $\forall \mathbf{x} \in V \setminus \{0\}, \Omega(\mathbf{x}, \mathbf{x}) > 0$ and $\Omega(\mathbf{0}, \mathbf{0}) = 0$.



8. Symmetric Matrices and Positive-Definite Matrices

Theorem: For a real-valued, finite-dimensional vector space V and an ordered basis B of V , it holds that $\langle ., . \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $A \in \mathbb{R}^{n \times n}$ with $\langle x, y \rangle = \hat{x}A^T\hat{y}$

An $n \times n$ matrix A is said to be symmetric if $A = A^T$

An $n \times n$ matrix A is said to be positive definite if for every non zero element x in $\mathbb{R}^n$, $x^T A x > 0$.

Problem 4



Consider an inner product space with an inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{A} \mathbf{y}$ defined with the help of matrix $\mathbf{A}$ defined below.

$$\mathbf{A} = \begin{bmatrix} 5.5 & -1.5 \\ -1.5 & 5.5 \end{bmatrix}$$

Check whether $\mathbf{A}$ is symmetric and positive definite.

Consider two vectors $\mathbf{a} = [1 \ 5]^T$ and $\mathbf{b} = [2 \ 7]^T$ in the inner product space.

- Find the distance $d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$ between vectors $\mathbf{a}$ and $\mathbf{b}$ in the above inner product space where $\|\cdot\|$ is the norm induced by the inner product.
- Find the angle between vectors $\mathbf{a}$ and $\mathbf{b}$ in the above inner product space.

Problem 4 – Solution

$$(a) \quad d(a, b) = \|a - b\| = c$$

$$d(a, b) = \sqrt{c^T A c}$$

$$[d(a, b)]^2 = c^T A c$$

$$(d(\mathbf{a}, \mathbf{b}))^2 = \begin{bmatrix} 1 & -2 & 5 & -7 \end{bmatrix} \begin{bmatrix} 5.5 & -1.5 \\ -1.5 & 5.5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 5 & -7 \end{bmatrix} = 21.5$$

$$d(\mathbf{a}, \mathbf{b}) = 4.63$$

$$(b) \quad \cos(\theta) = \frac{\mathbf{a}^T A \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{\mathbf{a}^T A \mathbf{b}}{\sqrt{\mathbf{a}^T A \mathbf{a}} \sqrt{\mathbf{b}^T A \mathbf{b}}}$$

$$\cos(\theta) = \frac{178}{\sqrt{128} \sqrt{249.5}} = 0.9960$$

$$\theta^0 = 5.1260$$

THANK YOU